When *Voyager 2* raced through the Neptune system in 1989, its cameras, spectrometers, particle and field detectors, and other instruments were feverishly examining the planet, its moons, and its rings. The planet itself, like its cousins Jupiter, Saturn, and Uranus, is a giant. Every planet is an Earthlike world at heart—but the four gas giants wear elaborate, cumbersome disguises. Jupiter and Saturn are great gas worlds with relatively small rocky and icy cores. But Uranus and Neptune are fundamentally rock and ice worlds swaddled in dense atmospheres that hide them from view.

Neptune is four times bigger than the Earth. When we look down on its cool, austere blueness, again we are seeing only atmosphere and clouds—no solid surface. Again, the atmosphere is made mainly of hydrogen and helium, with a little methane and traces of other hydrocarbons. There may also be some nitrogen. The bright clouds, which seem to be methane crystals, float above thick, deeper clouds of unknown composition. From the motion of the clouds we discovered fierce winds, approaching the local speed of sound. A Great Dark Spot was found, curiously at almost the same latitude as the Great Red Spot on Jupiter. The azure color seems appropriate for a planet named after the god of the sea.

Surrounding this dimly lit, chilly, stormy, remote world is—here also—a system of rings, each composed of innumerable orbiting objects ranging in size from the fine particles in cigarette smoke to small trucks. Like the rings of the other Jovian planets, those of Neptune seem to be evanescent—it is calculated that gravity and solar radiation will disrupt them in much less than the age of the Solar System. If they are destroyed quickly, we must see them only because they were made recently. But how can rings be made?

The biggest moon in the Neptune system is called Triton.* Nearly six of our days are required for it to orbit Neptune, which—alone among big moons in the Solar System—it does in the opposite direction to which its planet spins (clockwise if we say Neptune rotates counterclockwise). Triton has a nitrogen-rich atmosphere, somewhat similar to Titan's; but, because the air and haze are much thinner, we can see its surface. The landscapes are varied and splendid. This is a world of ices—nitrogen ice, methane ice,